

HIERO ECOSYSTEM

Career Intelligence, Smart Resume Compiler, Adaptive Voice Mock Interviewer & Placement Bridge

The comprehensive architectural, operational, and onboarding guide for engineers, AI researchers, and contributors joining the HIERO team.

- 🚀 Express Unified Gateway (Port 2816)
- 🌉 Bridge Portal (Port 2410 / 5050)
- 🧠 Groq LLaMA 3.3 70B & OpenRouter
- 🔊 Deepgram Nova-2 STT & Aura TTS
- 📄 22+ ATS Resume Templates (Puppeteer PDF)
- 🌐 IN 5-Language Regional Roadmaps (NEP 2020)
- 🗄️ MongoDB Atlas Cloud NoSQL

PRIMARY TARGET

Full-Stack & AI Team Onboarding

REPOSITORIES COVERED

Final-Hiero + Bidge-Portal

NATIONAL ALIGNMENT

Skill India • Digital India • NEP 2020

0. Executive Table of Contents

This onboarding document is organized into 11 logical modules designed to ramp up any new engineer or collaborator in under 60 minutes.

1. Executive Summary & Problem Statement — Employability crisis, NEP 2020 & Skill India	Page 2
2. High-Level System Architecture — Dual-repository design, port maps, and data flow	Page 3
3. Deep Dive: Final-Hiero (Student Platform) — Gateway, Auth, Analysis, Resume Compiler	Page 4
4. Deep Dive: AI Mock Interview & Voice Engine — Company Blueprints, Groq LLM, Deepgram	Page 6
5. Deep Dive: Bidge-Portal (Placement Hub) — TPO, Student, Recruiter workflows & CSV engine	Page 8
6. Mathematical Formulas & Scoring Algorithms — Fit Score, ATS Score, Interview 5-Pillar Score	Page 10
7. Database Schemas & Data Models — MongoDB Atlas Collections, Mongoose models	Page 11
8. Complete Tech Stack & Dependencies — Comprehensive library & package directory	Page 12
9. Environment Variables & Secrets Guide — Master .env configuration template	Page 13
10. Developer Quickstart & Local Setup Guide — Step-by-step local development setup	Page 14
11. Troubleshooting, Pitfalls & Pitch Guide — FAQ, debugging, and SIH pitch scripts	Page 15

1. Executive Summary & Problem Statement

 **The Core Mission of HIERO**

HIERO was engineered to solve the acute employability and placement mismatch facing over 1.5 million engineering and technical graduates annually in India, particularly across Tier-2 and Tier-3 institutions. It bridges the gap between academic transcripts and real-world industrial job descriptions (JDs).

1.1 The Employability Crisis in Numbers

- **80%+ of Technical Graduates** are deemed unemployable by top-tier product and service firms without 3–6 months of remedial training.
- **75% of Resumes** are rejected before human review by automated Applicant Tracking Systems (ATS) due to improper formatting, missing action verbs, and keyword mismatches.
- **Regional Language Barrier:** Students outside metropolitan hubs comprehend technical concepts significantly faster in regional languages, yet 95% of upskilling resources are exclusively in English.
- **Placement Office Inefficiencies:** Training & Placement Officers (TPOs) struggle with manual CSV tracking, disjointed screening, and lack of real-time candidate readiness visibility.

1.2 Alignment with National Policy Frameworks

IN Skill India & Digital India

Democratizes enterprise-grade resume building, ATS compliance testing, and AI technical interview practice at zero cost to students nationwide.

ALIGNED

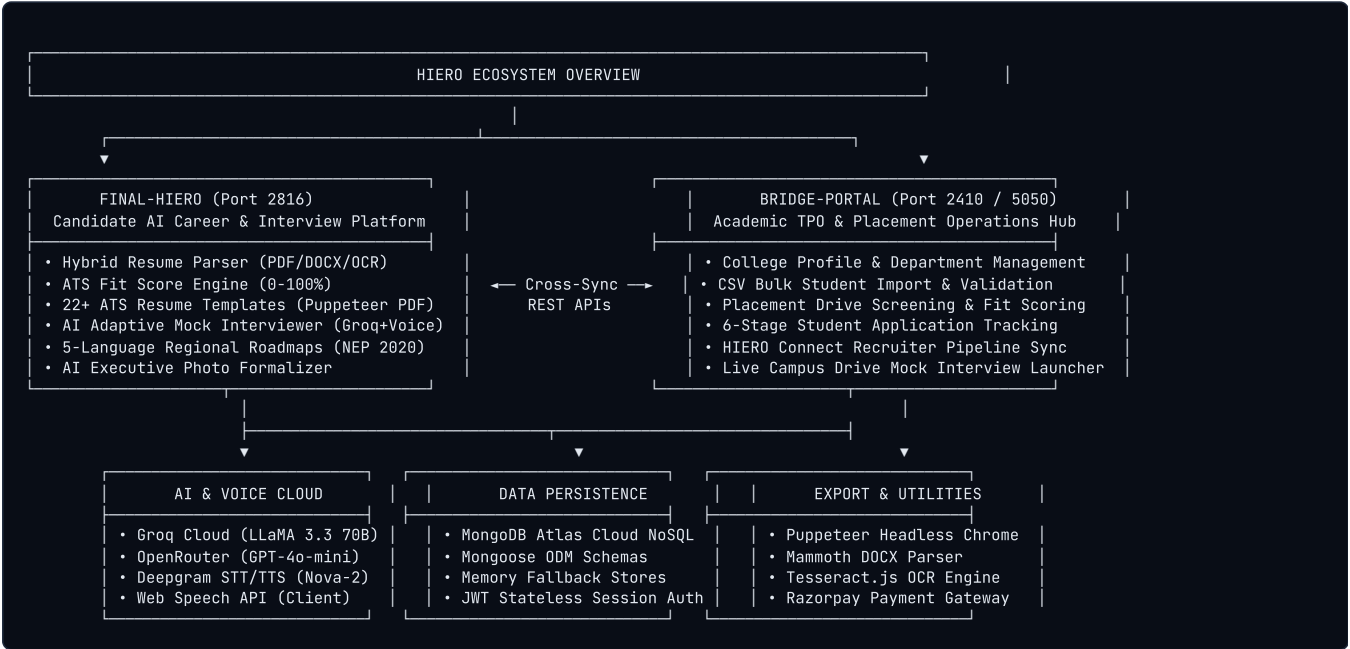
 **National Education Policy (NEP 2020)**

Promotes mother-tongue learning by generating personalized YouTube technical learning playlists in **5 Indian languages:** English, Hindi, Telugu, Tamil, and Kannada.

MULTILINGUAL

2. High-Level System Architecture

The HIERO ecosystem comprises two tightly integrated platforms: **Final-Hiero** (the candidate-facing AI career companion) and **Bidge-Portal** (the academic institution & recruiter placement bridge).



2.1 Port & Service Allocation Map

Service Name	Port	Main File	Primary Purpose
Final-Hiero Unified Gateway	2816	Final-Hiero/gateway.js	Single unified express entrypoint serving frontend UI, REST API, Auth, Resume Builder, & AI Analysis.
Bridge-Portal Unified Gateway	2410	Bidge-Portal/gateway.js	Unified portal serving Academic TPO UI, Student Drive application UI, and Bridge REST endpoints.
Bridge Standalone REST API	5050	Bidge-Portal/backend/src/server.js	Standalone backend REST server for Bridge Portal when running detached from frontend.
MongoDB Atlas Cloud	27017 (Cloud)	mongodb+srv:// ...	Centralized NoSQL cloud database storing users, resumes, interview sessions, reviews, and tracking.

3. Deep Dive: Final-Hiero (Student Platform)

3.1 Unified Gateway Architecture (gateway.js)

Historically, the project had multiple disjoint microservices running on ports 8080, 5001, and 5003. To eliminate cross-origin (CORS) bugs, cookie loss, and deployment overhead, all services were consolidated into a **Single Unified Express Gateway** running on Port 2816.

- **Static Asset Serving:** Serves landing pages, dashboard, resume builder, mock interview UI, and company interview preparation pages directly with proper caching headers.
- **DNS SRV Fix for Windows:** Overrides Node.js default DNS resolvers using Google (8.8.8.8 , 8.8.4.4) and Cloudflare (1.1.1.1) to guarantee MongoDB Atlas SRV URI resolution on Windows machines.
- **Dual .env Cascading:** Automatically loads variables from login-system/.env and cascades root .env overrides gracefully.
- **Memory Resilience:** Implements in-memory fallbacks for session states and user data when MongoDB is offline during offline hackathon demos.

3.2 Authentication & Security Subsystem (routes/auth.js)

Auth Method	Strategy / Library	Flow Description
Google OAuth 2.0	passport-google-oauth20	One-click sign-in via Google accounts. Automatically retrieves name, email, and avatar, creates/updates user in DB, and issues signed JWT.
GitHub OAuth	passport-github2	Developer-centric authentication capturing GitHub username, public repos, and verified email.
Email / Password	bcryptjs (10 salt rounds)	Traditional credentials with email verification support via nodemailer .
Stateless JWT	jsonwebtoken	Bearer token stored in client localStorage , validated across all protected API routes.

3.3 Hybrid Resume Parser & Text Extraction (routes/analysis.js)

Resumes uploaded by students arrive in diverse formats: digital vector PDFs, Word DOCX files, and scanned image-only PDFs. HIERO utilizes a 3-tier fallback extraction pipeline:

Tier 1 & 2: Digital Vector & DOCX Uses pdf-parse for high-speed text stream extraction from vector PDFs. Uses mammoth.js for extracting clean raw text and structure from Microsoft Word DOCX files.	Tier 3: OCR Engine If extracted text length is < 50 characters (indicating a scanned or rasterized image PDF), the system triggers Tesseract.js optical character recognition on page images.
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3.4 22+ ATS Resume Template & Puppeteer PDF Compiler

HIERO features 22+ production-grade ATS resume templates categorized into **Executive, Modern, Creative, Tech-First, and Academic** styles:

- **Single-Column ATS Compliant:** Built specifically to pass enterprise ATS scanners (Workday, Greenhouse, Taleo) with 100% parse accuracy.
- **Dynamic In-Browser Preview:** Instant DOM manipulation allows real-time live editing and instant layout switching without page reloading.
- **Puppeteer Vector PDF Rendering:** Headless Chromium compiles DOM into crystal-clear vector PDFs with selectable text, clickable hyperlinks, and exact 0.5-inch margins.
- **DOCX Word Export:** Generates fully editable Word documents via html-to-docx for users needing offline desktop tweaking.

4. AI Mock Interview & Voice Engine

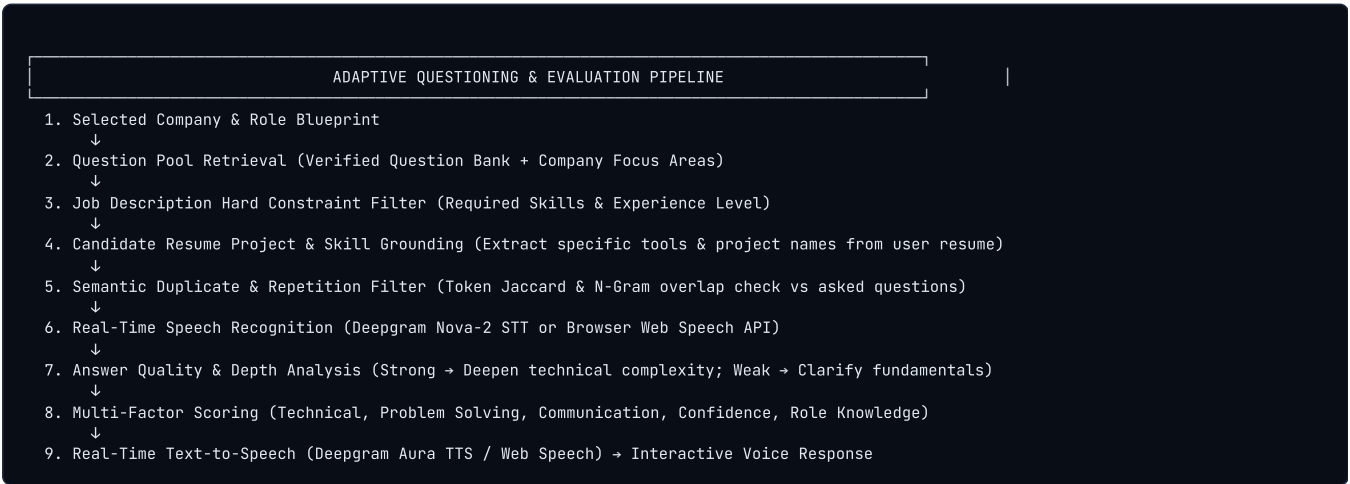
The AI Mock Interview Engine (`routes/interview.js` and `routes/interviewEngine.js`) simulates realistic, high-pressure technical and behavioral interviews tailored to top tech companies.

4.1 Company Blueprint System

Each company profile in HIERO enforces exact hiring bars, signature interview phases, and domain focus areas:

Company	Hiring Bar	Category Weights (Tech / SysDesign / Resume / Behavioral)	Primary Culture & Focus Areas
Google	9.5 / 10	35% Tech 35% SysDesign 20% Resume 10% Behavioral	Algorithmic rigor, distributed systems scalability, Googliness, trade-off analysis.
Microsoft	9.2 / 10	35% Tech 30% SysDesign 25% Resume 10% Behavioral	Growth mindset, Azure cloud reliability, OOP clean code, API idempotency.
Amazon	9.3 / 10	30% Tech 30% SysDesign 25% Resume 15% Behavioral	16 Leadership Principles (Customer Obsession, Ownership, STAR methodology).
Meta	9.4 / 10	40% Tech 35% SysDesign 15% Resume 10% Behavioral	High-concurrency systems, social graphs, caching/sharding, rapid coding speed.
Apple	9.4 / 10	40% Tech 25% SysDesign 25% Resume 10% Behavioral	Precision engineering, low-latency memory management, user privacy & craftsmanship.
Databricks	9.6 / 10	45% Tech 35% SysDesign 15% Resume 5% Behavioral	Distributed computing internals, Apache Spark, query engines, storage layers.

4.2 Multi-Stage Adaptive Interview Pipeline



4.3 Speech-to-Text & Text-to-Speech Architecture (`services/deepgramService.js`)

- Primary Engine (Deepgram Nova-2 & Aura):** Provides ultra-low latency transcription (< 250ms) and natural human-like voice synthesis. Audio data is streamed as base64 or audio buffer chunks.
- Fallback Engine (Web Speech API):** Seamlessly executes directly in the browser if Deepgram API quotas or network limits occur, ensuring uninterrupted interview simulations during offline judging.
- Full Session Audio/Video Recording:** Collects WebM audio/video chunks via `MediaRecorder` for retrospective review and body language analytics.

5. Deep Dive: Bidge–Portal (Academic Placement Hub)

HIERO Bridge is the dedicated academic and placement management portal connecting university Training & Placement Offices (TPOs), final-year students, and corporate recruiters.

5.1 Four Specialized Role Modules

1. College Administration ADMIN - Configure College profile (Name, Code, Tier, Location, Batch). - Manage academic departments (CSE, AIML, ECE, MECH). Bulk CSV Student Import: Real-time parser with duplicate detection, regex email checks, and GPA validation.	2. Placement Coordinator (TPO) TPO HUB - Live placement analytics (Total students, active drives, selections). Automated Candidate Screening: Filters candidates by CGPA cutoffs, active backlogs, and required technical skills. 1-Click candidate shortlisting with coordinator notes.
3. Final-Year Student Gateway STUDENT - Verified student academic profile with locked credentials. Drive Eligibility Engine: Instant real-time eligibility computation with specific reasons if disqualified. 1-Click application submission with PDF resume attachment. 6-stage visual application progress timeline.	4. Recruiter & HIERO Connect RECRUITER - Publish campus hiring drives (Role, CTC, Criteria, Skills). - Direct candidate pipeline sync from College TPO shortlists. - Update candidate stages (Shortlisted, Interview, Selected, Rejected).

5.2 CSV Bulk Import & Validation Engine

The bulk import pipeline (`POST /api/admin/students/bulk-import`) processes multi-column CSVs containing hundreds of student records. It outputs an immediate audit report:

- Validation Checks:** Validates registration number uniqueness, CGPA boundaries (`0.0 - 10.0`), department code mapping, and email format.
- Duplicate Detection:** Compares uploaded rows against existing student database records to prevent redundant entries.
- Staged Confirmation:** Coordinators review valid vs invalid rows before executing the database commit (`POST /api/admin/students/confirm-import`).

5.3 Bridge–Portal REST API Contract

Method	Endpoint	Description	Key Parameters / Payload
POST	<code>/api/auth/login</code>	Role-based authentication	<code>{ role: "admin" "coordinator" "student", email }</code>
GET	<code>/api/coordinator/metrics</code>	Aggregated placement KPI counters	Returns total students, active drives, shortlisted count.
POST	<code>/api/coordinator/opportunities/:id/screen</code>	Automated candidate filter & fit scoring	<code>{ minCGPA, depts, requiredSkills, minProjects }</code>
POST	<code>/api/coordinator/opportunities/:id/shortlist</code>	Endorse candidate list to recruiter	<code>{ studentIds: ["STU-001", "STU-002"], notes }</code>
GET	<code>/api/student/drives</code>	List eligible placement drives	<code>?studentId=STU-001</code> (computes <code>isEligible</code> flag)
POST	<code>/api/student/apply</code>	Submit campus drive application	<code>multipart/form-data (oppId, studentId, resume.pdf)</code>
POST	<code>/api/interview/evaluate</code>	Evaluate domain mock interview	<code>{ appId, oppId, answers: [...] }</code> → 5-pillar scorecard

6. Mathematical Formulas & Scoring Algorithms

6.1 Overall Resume-to-JD Match Score (0 – 100)

The match score quantifies how well a candidate's credentials align with a specific Job Description:

Match Score = Min(100, SkillPoints + ProjectPoints + ExperiencePoints + EducationPoints + ExtraBonus)

1. Skill Points (Max 40 Pts):
SkillPoints = (Count(Matched_JD_Skills) / Total_JD_Skills) * 40

2. Project Points (Max 25 Pts):
ProjectPoints = Base_Project_Points (15) + (JD_Project_Skill_Alignment_Ratio * 10)

3. Experience Points (Max 15 Pts):
ExperiencePoints = Base_Exp (9) + Min(6, Years_Of_Experience * 2)

4. Education Points (Max 9 Pts):
EducationPoints = 9 (if accredited Degree B.Tech/M.Tech/MCA/B.Sc is detected)

5. Extra Technical Bonus (Max 10 Pts):
ExtraBonus = Min(10, Count(Extra_Technical_Skills_Not_In_JD) * 1.5)

6.2 ATS Compatibility & Formatting Score (0 – 100)

ATS Score = (0.30 * Keyword_Density) + (0.20 * Standard_Structure) + (0.20 * Quantifiable_Impact) + (0.20 * Readability_Metric) + (0.10 * Action_Verbs_Ratio)

Where:

- Standard_Structure (20 Pts): Presence of 5 essential headers [Summary, Experience, Projects, Skills, Education]
- Action_Verbs_Ratio (10 Pts): Detection of active power verbs (e.g., 'Architected', 'Optimized', 'Deployed')
- Quantifiable_Impact (20 Pts): Scans for numerical metrics (e.g., '40% latency reduction', '10k+ MAU')
- Readability_Metric (20 Pts): Optimal word density (8 to 20 words per bullet line)
- Formatting_Bounds (20 Pts): Optimal overall word count between 350 and 800 words

6.3 AI Voice Interview 5–Pillar Scorecard (0 – 100)

Evaluates answers across 5 independent vectors using Groq LLM parameter analysis:

Pillar	Weight	Evaluation Criteria
1. Technical Accuracy	30%	Correctness of algorithms, architectural correctness, edge-case coverage, and complexity analysis.
2. Problem Solving	25%	Step-by-step reasoning, decomposition of ambiguous constraints, trade-off evaluation.
3. Communication Clarity	20%	Conciseness, structured articulation, logical transitions, vocabulary precision.
4. Confidence & Delivery	15%	Pace, minimal filler words (um, ah), directness, intellectual conviction.
5. Role Knowledge & Depth	10%	Familiarity with industry production patterns, modern toolchains, and company-specific values.

7. Database Schemas & Data Models

HIERO utilizes **MongoDB Atlas** with Mongoose ODM schemas. In addition, an in-memory Map structure provides high-availability fallback.

7.1 Core MongoDB Models (`models/`)

User.js

AUTH & PROFILE

```
{
  name: { type: String, required: true },
  email: { type: String, unique: true },
  password: { type: String }, // bcrypt hashed
  picture: { type: String },
  role: { type: String, default: 'student' },
  college: { type: String },
  department: { type: String },
  verified: { type: Boolean, default: false },
  createdAt: { type: Date, default: Date.now }
}
```

Resume.js

PARSED DATA

```
{
  userId: { type: String, index: true },
  data: {
    personal: { name, email, phone, links },
    education: [ { school, degree, year, gpa } ],
    experience: [ { company, role, bullets } ],
    projects: [ { name, tech, bullets, link } ],
    skills: { technical: [], soft: [] }
  },
  lastUpdated: { type: Date, default: Date.now }
}
```

InterviewSession.js

MOCK SESSIONS

```
{
  sessionId: { type: String, unique: true },
  userId: { type: String, index: true },
  companyId: { type: String },
  roleId: { type: String },
  status: { type: String, enum: ['active', 'completed'] },
  questions: [ {
    qId: String, text: String, category: String,
    answerText: String, score: Number, feedback: String
  } ],
  scorecard: {
    overallScore: Number, technicalScore: Number,
    communicationScore: Number, weakAreas: [String]
  },
  createdAt: { type: Date, default: Date.now }
}
```

Review.js

FEEDBACK & RATINGS

```
{
  userId: { type: String },
  name: { type: String },
  email: { type: String },
  rating: { type: Number, min: 1, max: 5 },
  comment: { type: String },
  role: { type: String },
  featureUsed: { type: String },
  createdAt: { type: Date, default: Date.now }
}
```

8. Tech Stack & Dependencies Index

Category	Package / Service	Version	Role in Ecosystem
Runtime & Framework	express	^4.22.1	Core HTTP web application framework for Unified Gateway on Port 2816 and Bridge.
AI Intelligence	groq-sdk	^1.1.1	Ultra-fast inference client executing LLaMA 3.3 70B for resume analysis & mock interviews.
AI Fallback	@google/generative-ai	^0.24.1	Gemini Pro API integration for secondary parsing and coding challenge generation.
PDF Compilation	puppeteer	^24.37.5	Headless Chrome automation to compile live HTML/CSS resume templates into vector PDFs.
Document Extraction	pdf-parse , mammoth	^1.1.1 , ^1.11.0	Text stream parsers for uploaded vector PDFs and Microsoft Word DOCX files.
OCR Vision	tesseract.js	^5.1.1	Optical Character Recognition for scanned images and non-selectable PDF resumes.
Voice & Audio	deepgramService.js	REST / WS	Deepgram Nova-2 speech-to-text and Aura natural text-to-speech engine.
Database & ODM	mongoose	^7.6.3	MongoDB object modeling tool for asynchronous schema validation and queries.
Auth & Security	passport , jsonwebtoken , bcryptjs	Latest	OAuth 2.0 Google/GitHub strategies, stateless JWT token signing, and password hashing.
Payments	razorpay	^2.9.2	Payment gateway integration for premium resume templates and mock interview credits.

9. Environment Variables & Configuration

Create a `.env` file in both the `Final-Hiero` root and `Bidge-Portal` root. Below is the master reference template:

```
# =====
# FINAL-HIERO MASTER ENVIRONMENT CONFIGURATION (.env)
# =====
PORT=2816
NODE_ENV=development
PUBLIC_URL=http://localhost:2816

# — MONGODB ATLAS —
MONGODB_URI=mongodb+srv://<username>:<password>@cluster0.mongodb.net/hiero_db?retryWrites=true&w=majority

# — AUTHENTICATION & SECRETS —
JWT_SECRET=hiero_jwt_production_secret_key_2026_super_secure
GOOGLE_CLIENT_ID=your_google_oauth_client_id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=your_google_oauth_client_secret
GITHUB_CLIENT_ID=your_github_client_id
GITHUB_CLIENT_SECRET=your_github_client_secret

# — ARTIFICIAL INTELLIGENCE & LLMS —
GROQ_API_KEY=gsk_your_groq_api_key_here
OPENROUTER_API_KEY=sk-or-v1-your_openrouter_key_here
GEMINI_API_KEY=AiZaSy_your_gemini_api_key_here

# — VOICE & SPEECH SYNTHESIS (DEEPGRAM) —
DEEPGRAM_API_KEY=your_deepgram_api_key_here

# — YOUTUBE DATA API (5-LANGUAGE ROADMAPS) —
YOUTUBE_API_KEY=AiZaSy_your_youtube_v3_api_key_here

# — PAYMENTS (RAZORPAY) —
RAZORPAY_KEY_ID=rzp_test_your_razorpay_key
RAZORPAY_KEY_SECRET=your_razorpay_secret_key
```

10. Developer Quickstart & Local Setup

Step 1: Clone and Install Dependencies

```
# 1. Clone repositories
git clone https://github.com/jaswanthkumar-2816/Final-Hiero.git
git clone https://github.com/jaswanthkumar-2816/Bidge-Portal.git

# 2. Install Final-Hiero dependencies
cd Final-Hiero
npm install

# 3. Install Bridge-Portal dependencies
cd ../Bidge-Portal
npm install
```

Step 2: Start the Development Servers

Terminal 1: Start Final-Hiero

PORT 2816

```
cd Final-Hiero
node gateway.js
# UI: http://localhost:2816
# Dashboard: http://localhost:2816/dashboard
# Mock Interview: http://localhost:2816/mock-interview
```

Terminal 2: Start Bridge Portal

PORT 2410

```
cd Bidge-Portal
node gateway.js
# Academic Portal: http://localhost:2410/academic-portal.html
# Student Apply: http://localhost:2410/apply
# REST API Base: http://localhost:2410/api
```

11. Troubleshooting & Team Pitch Master Guide

11.1 Common Gotchas & Instant Fixes

Issue / Symptom	Root Cause	Immediate Resolution
MongoDB Atlas SRV DNS Timeout on Windows	Windows DNS ISP blocks SRV record lookups.	Ensure <code>dns.setServers(['8.8.8.8', '1.1.1.1'])</code> runs before Mongoose connects (already in <code>gateway.js</code>).
Puppeteer Chrome Launch Error on Linux/Docker	Missing sandbox flags or shared libraries.	Launch Puppeteer with <code>args: ['--no-sandbox', '--disable-setuid-sandbox']</code> .
CORS Blocked on API Calls	Frontend requesting mismatched port.	Use the unified gateway URLs (<code>http://localhost:2816</code>) directly rather than cross-calling ports.
OAuth Redirect URI Mismatch	Google/GitHub Console callback URL mismatch.	Configure authorized redirect URI in Google Console to <code>http://localhost:2816/auth/google/callback</code> .

11.2 Master Pitch Scripts for Juries and Stakeholders

⚡ 30-Second Elevator Pitch

"HIERO is an AI-powered career ecosystem that democratizes placement readiness for millions of Indian students. It evaluates resume-to-job fit, generates 5-language regional upskilling roadmaps in Hindi, Telugu, Tamil, Kannada, and English aligned with NEP 2020, builds 22+ ATS resumes, and conducts adaptive AI voice mock interviews."

🏆 3-Minute Comprehensive Demo Script

- Minute 1 (Problem & National Impact):** Highlight that 80%+ of Indian graduates face placement rejections due to skill gaps and ATS filters. Mention Tier-2/3 language barriers and NEP 2020 alignment.
- Minute 2 (Live Resume Analysis & 5-Language Learning):** Upload a sample resume + JD live. Show instant 0-100 fit score, missing skills breakdown, and the auto-generated regional YouTube playlist. Switch between 22 ATS templates and export high-res PDF.
- Minute 3 (AI Voice Interview & Placement Bridge):** Trigger the AI Mock Interviewer. Answer a live question via microphone and demonstrate real-time scoring. Switch to Bridge Portal to show how TPOs screen students and export shortlisted batches to recruiters.